

DDCET 2026 — PREDICTED QUESTION PAPER

Gujarat Technological University · Engineering Entrance

Questions: 100 | Total Marks: 200 | Time: 150 Minutes | +2 per correct | –0.5 negative

SECTION 1 — Basics of Science and Engineering (Q.01 to Q.50 · 100 Marks)

UNITS & MEASUREMENT — Q.01 TO Q.06

Q.01 The pitch of a micrometer screw gauge is 1 mm and the number of divisions on its circular scale is 100. What is its least count?

- A.** 0.1 mm
- B.** 0.01 mm
- C.** 0.001 mm
- D.** 1 mm

Q.02 The SI unit of pressure is _____.

- A.** Newton
- B.** Pascal
- C.** Joule
- D.** Bar

Q.03 1 Newton = _____ dyne.

- A.** 10^3
- B.** 10^4
- C.** 10^5
- D.** 10^7

Q.04 The dimensional formula of force is _____.

- A.** $[MLT^{-1}]$
- B.** $[MLT^{-2}]$
- C.** $[ML^2T^{-2}]$
- D.** $[ML^{-1}T^{-2}]$

Q.05 The following readings of a physical quantity are noted: 5.00, 5.02, 4.98, 5.01, 4.99. What is the mean absolute error?

- A.** 0.012
- B.** 0.0012
- C.** 0.12
- D.** 1.2

Q.06 If length $L = 5.0 \pm 0.1$ cm and breadth $B = 4.0 \pm 0.1$ cm, the percentage error in area $A = L \times B$ is approximately _____.

- A.** 2%
- B.** 2.5%
- C.** 4.5%
- D.** 5%

CLASSICAL MECHANICS — Q.07 TO Q.12

Q.07 A car of mass 1000 kg starts from rest and reaches 20 m/s in 10 seconds. What force is applied?

- A.** 1000 N
- B.** 2000 N
- C.** 500 N
- D.** 200 N

Q.08 A body of mass 4 kg moves with velocity 5 m/s. Its kinetic energy is _____.

- A.** 20 J
- B.** 50 J
- C.** 40 J
- D.** 100 J

Q.09 A body moves in a circle of radius 4 m with uniform speed 8 m/s. Its centripetal acceleration is _____.

- A.** 2 m/s²
- B.** 32 m/s²
- C.** 16 m/s²
- D.** 64 m/s²

Q.10 If the velocity of a body is doubled, its kinetic energy becomes _____.

- A.** Double
- B.** Half
- C.** Four times
- D.** Eight times

Q.11 A force of 100 N acts on a body through a displacement of 5 m at angle 60° to the direction of force. Work done is _____.

- A.** 500 J
- B.** 433 J
- C.** 250 J
- D.** 0 J

Q.12 Centrifugal force is a _____ force.

- A.** Real force acting outward
- B.** Pseudo (fictitious) force acting outward
- C.** Real force acting inward
- D.** Nuclear force

ELECTRIC CURRENT — Q.13 TO Q.18

Q.13 Three resistors 3 Ω, 6 Ω and 9 Ω are connected in series. Equivalent resistance is _____.

- A.** 18 Ω
- B.** 1.6 Ω
- C.** 6 Ω
- D.** 9 Ω

Q.14 According to Coulomb's law, the electrostatic force between two charges is _____.

- A.** Directly proportional to the product of charges and directly proportional to square of distance
- B.** Directly proportional to the product of charges and inversely proportional to square of distance
- C.** Inversely proportional to product of charges and directly proportional to distance
- D.** Directly proportional to distance and product of charges

Q.15 The unit of electric field intensity is _____.

- A.** N·m
- B.** N/C
- C.** C/N
- D.** V·m

Q.16 A charge of 12 μC is placed on a capacitor which develops a potential difference of 3 V. The capacitance is _____.

- A.** 36 μF
- B.** 4 μF
- C.** 0.25 μF
- D.** 4 F

Q.17 The resistivity of copper is $1.72 \times 10^{-8} \Omega \cdot \text{m}$. Resistance of a 2 m long copper wire with cross-sectional area 10^{-6} m^2 is _____.

- A.** 0.0344 Ω
- B.** 0.344 Ω
- C.** 3.44 Ω
- D.** 34.4 Ω

Q.18 Which of the following is NOT a property of electric flux?

- A.** It depends on the electric field intensity
- B.** It depends on the area of the surface
- C.** It depends on the angle between surface and field
- D.** It depends on the temperature of the conductor

HEAT & THERMOMETRY — Q.19 TO Q.24

Q.19 Convert 300 K to Celsius scale.

- A.** 573°C
- B.** 27°C
- C.** -27°C
- D.** 300°C

Q.20 A piece of iron (mass 200 g, specific heat 0.12 cal/g·°C) is heated from 20°C to 120°C. Heat absorbed is _____.

- A.** 2400 cal
- B.** 240 cal
- C.** 24 cal
- D.** 24000 cal

Q.21 The process of heat transfer that can occur through a vacuum is _____.

- A.** Conduction
- B.** Convection
- C.** Radiation
- D.** Both A and B

Q.22 A steel rod of length 2 m has coefficient of linear expansion $\alpha = 1.2 \times 10^{-5} / ^\circ\text{C}$. If heated from 0°C to 100°C, the increase in length is _____.

- A.** 2.4×10^{-2} m
- B.** 2.4×10^{-3} m
- C.** 2.4×10^{-4} m
- D.** 1.2×10^{-3} m

Q.23 At what temperature does the Fahrenheit reading become twice the Celsius reading?

- A.** 160°C
- B.** 80°C
- C.** -40°C
- D.** 40°C

Q.24 The mathematical formula for specific heat is _____.

- A.** $C = Q / m\Delta T$
- B.** $Q = C / m\Delta T$
- C.** $C = m\Delta T / Q$
- D.** $Q = m / C\Delta T$

WAVE MOTION, OPTICS & ACOUSTICS — Q.25 TO Q.30

Q.25 Optical fibre works on the principle of _____.

- A.** Diffraction
- B.** Polarization
- C.** Total internal reflection
- D.** Refraction

Q.26 A sound wave has frequency 340 Hz and wavelength 1 m. The speed of the sound wave is _____.

- A.** 340 m/s
- B.** 34 m/s
- C.** 3400 m/s
- D.** 3.4 m/s

Q.27 The critical angle for glass-air interface (refractive index of glass = 1.5) is approximately _____.

- A.** 30°
- B.** 48.6°
- C.** 41.8°
- D.** 60°

Q.28 An auditorium has volume 8000 m^3 and total absorption by surfaces is 1600 O.W.U. Find the reverberation time using Sabine's formula.

- A.** 0.825 s
- B.** 8.25 s
- C.** 82.5 s
- D.** 0.0825 s

Q.29 Which of the following is NOT a characteristic of LASER?

- A.** Coherent
- B.** Monochromatic
- C.** Divergent
- D.** High intensity

Q.30 Total internal reflection occurs when light travels from _____.

- A.** Rarer to denser medium at any angle
- B.** Denser to rarer medium at angle greater than critical angle
- C.** Rarer to denser medium at critical angle
- D.** Denser to rarer medium at angle less than critical angle

CHEMICAL REACTIONS & EQUATIONS — Q.31 TO Q.33

Q.31 The reaction $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ is an example of _____.

- A.** Combination reaction
- B.** Displacement reaction
- C.** Double displacement reaction
- D.** Decomposition reaction

Q.32 Which of the following is the correctly balanced equation for the reaction of zinc with sulphuric acid?

- A.** $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$
- B.** $2\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$
- C.** $\text{Zn} + 2\text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + 2\text{H}_2$
- D.** $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + 2\text{H}$

Q.33 Ferrous sulphate crystals ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) on heating lose their water of crystallisation and change colour from green to white. This is an example of _____.

- A.** Combination reaction
- B.** Displacement reaction
- C.** Decomposition reaction
- D.** Double displacement reaction

ACIDS, BASES & SALTS — Q.34 TO Q.37

Q.34 The pH of a neutral solution at 25°C is _____.

- A.** 0
- B.** 7
- C.** 14
- D.** 1

Q.35 Metal oxide + Acid $\rightarrow$ _____

- A.** Base + Water
- B.** Salt + Water
- C.** Acid + Salt
- D.** Salt + Oxygen

Q.36 The approximate pH of human blood is _____.

- A.** 7.0
- B.** 6.8
- C.** 7.4
- D.** 8.0

Q.37 The outermost shell of fluorine contains how many electrons?

- A.** 9
- B.** 7
- C.** 2
- D.** 5

METALS & NON-METALS — Q.38 TO Q.40

Q.38 The process of depositing a layer of zinc on iron to prevent corrosion is called _____.

- A.** Tinning
- B.** Galvanizing
- C.** Electroplating
- D.** Anodizing

Q.39 Which non-metal is a good conductor of electricity?

- A.** Sulphur
- B.** Phosphorus
- C.** Graphite
- D.** Diamond

Q.40 The chemical formula of rust is _____.

- A.** Fe_2O_3
- B.** Fe_3O_4
- C.** $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- D.** FeO

COMPUTER PRACTICE — Q.41 TO Q.45

Q.41 Which of the following is NOT an Operating System?

- A.** UNIX
- B.** MS-DOS
- C.** LINUX
- D.** Microsoft Office

Q.42 The full form of HTML is _____.

- A.** Hyper Text Markup Language
- B.** High Text Markup Language
- C.** Hyper Transfer Markup Language
- D.** Hyper Text Machine Language

Q.43 Which of the following is NOT a valid data type in MS-Excel?

- A.** Number
- B.** Character
- C.** Label
- D.** Date/Time

Q.44 CPU consists of _____.

- A.** ALU + CU
- B.** ROM + ALU
- C.** RAM + ROM
- D.** RAM + CU

Q.45 The maximum zoom percentage in MS-PowerPoint is _____.

- A.** 400%
- B.** 300%
- C.** 200%
- D.** 100%

ENVIRONMENTAL SCIENCES — Q.46 TO Q.50

Q.46 Acid rain is primarily caused by which pollutants?

- A.** CO_2 and O_3
- B.** SO_2 and NO_x
- C.** CH_4 and CO
- D.** CFC and N_2O

Q.47 CFCs are not recommended in refrigerators because they _____.

- A.** Increase temperature
- B.** Affect environment
- C.** Deplete the ozone layer
- D.** Affect human body directly

Q.48 The approximate efficiency range of modern solar photovoltaic (PV) panels for converting sunlight into electricity is _____.

- A.** 5-10%
- B.** 15-20%
- C.** 40-50%
- D.** 70-80%

Q.49 Which pollutant is commonly associated with 'Blue Baby Syndrome' in infants?

A. Lead

B. Arsenic

C. Mercury

D. Nitrate

Q.50 Which of the following is NOT a renewable source of energy?

A. Solar energy

B. Wind energy

C. Coal

D. Tidal energy

SECTION 2 — Aptitude Test: Mathematics + Soft Skills (Q.51 to Q.100 · 100 Marks)

DETERMINANTS & MATRICES — Q.51 TO Q.54

Q.51 If $A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$, then $|A| =$ _____.

A. 10

B. 14

C. -10

D. -14

Q.52 If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then $\text{adj}(A) =$ _____.

A. $\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$

C. $\begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix}$

Q.53 If $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, then $A^{-1} =$ _____.

A. $(1/5)\begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$

B. $(1/5)\begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$

C. $(1/5)\begin{bmatrix} -4 & 3 \\ 1 & -2 \end{bmatrix}$

D. $(1/5)\begin{bmatrix} 2 & -3 \\ -1 & 4 \end{bmatrix}$

Q.54 If $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 0 \end{bmatrix}$, then $|A| =$ _____.

A. -3

B. 3

C. -4

D. 4

TRIGONOMETRY — Q.55 TO Q.57

Q.55 $\sin(120^\circ) =$ _____.

A. $1/2$

B. $\sqrt{3}/2$

C. $-\sqrt{3}/2$

D. $-1/2$

Q.56 If $\tan \theta = 3/4$ and θ is in the first quadrant, then $\sin \theta =$ _____.

A. $3/5$

B. $4/5$

C. $3/4$

D. $4/3$

Q.57 $\cos^{-1}(\sqrt{3}/2) =$ _____.

A. $\pi/2$

B. $\pi/6$

C. $\pi/3$

D. $\pi/4$

VECTORS — Q.58 TO Q.59

Q.58 The magnitude of vector $A = 2i + 3j + 6k$ is _____.

A. 5

B. 7

C. $\sqrt{49}$

D. 11

Q.59 If vectors $(2, 3, a)$ and $(3, -1, 4)$ are perpendicular, then $a =$ _____.

A. $3/4$

B. $-3/4$

C. $4/3$

D. $-4/3$

COORDINATE GEOMETRY — Q.60 TO Q.61

Q.60 The slope of the line $3x - 4y + 8 = 0$ is _____.

A. $3/4$

B. $-3/4$

C. $4/3$

D. $-4/3$

Q.61 The equation of a circle with centre $(2, -3)$ and radius 5 is _____.

A. $(x+2)^2 + (y-3)^2 = 25$

B. $(x-2)^2 + (y+3)^2 = 25$

C. $(x-2)^2 + (y+3)^2 = 5$

D. $(x+2)^2 + (y+3)^2 = 25$

FUNCTION & LIMIT — Q.62 TO Q.64

Q.62 $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2) =$ _____.

A. 0

B. 2

C. 4

D. ∞

Q.63 $\lim_{x \rightarrow 0} \sin(3x)/x =$ _____.

A. 0

B. 1

C. 3

D. $1/3$

Q.64 $\lim_{x \rightarrow 0} (10^x - 5^x)/x =$ _____.

A. $\ln 2$

B. $\ln 10$

C. $\ln 5$

D. $(1/2) \cdot \ln 2$

DIFFERENTIATION & APPLICATIONS — Q.65 TO Q.68

Q.65 $d/dx(\sin x + \cos x) =$ _____.

A. $\cos x + \sin x$

B. $\cos x - \sin x$

C. $-\cos x + \sin x$

D. $-\cos x - \sin x$

Q.66 $d/dx(\cos^2 x) =$ _____.

A. $\sin 2x$

B. $-\sin 2x$

C. $\cos 2x$

D. $-\cos 2x$

Q.67 If $x = \operatorname{cosec} \theta$ and $y = \cot \theta$, then $dy/dx =$ _____.

A. $\sec \theta$

B. $\operatorname{cosec} \theta$

C. $\sin \theta$

D. $\cos \theta$

Q.68 The necessary condition for $f(x)$ to have a minimum at $x = a$ is _____.

A. $f'(a) > 0$

B. $f''(a) < 0$

C. $f'(a) < 0$

D. $f''(a) > 0$

INTEGRATION — Q.69 TO Q.72

Q.69 integral of $\cos(10x + 17)$ dx = _____.

- A.** $10 \sin(10x+17) + C$ **B.** $-10 \sin(10x+17) + C$
C. $(1/10) \sin(10x+17) + C$ **D.** $-(1/10) \sin(10x+17) + C$

Q.70 Definite integral from -1 to 1 of $\sin^5 x \cdot \cos^8 x$ dx = _____.

- A.** -1 **B.** 0
C. $1/2$ **D.** $1/4$

Q.71 integral of $e^x(\cot x - \operatorname{cosec}^2 x)$ dx = _____.

- A.** $e^x \cot x + C$ **B.** $e^x \operatorname{cosec}^2 x + C$
C. $-e^x \cot x + C$ **D.** $e^x(\cot x - \operatorname{cosec} x) + C$

Q.72 integral of $3x/(x^2 + 6x + 40)$ dx = _____.

- A.** $(3/2) \log|x^2+6x+40| + C$ **B.** $(1/2) \log|x^2+6x+40| + C$
C. $3 \log|x^2+6x+40| + C$ **D.** $\log|x^2+6x+40| + C$

LOGARITHM — Q.73 TO Q.74

Q.73 $\log_2(32) =$ _____.

- A.** 3 **B.** 4
C. 5 **D.** 6

Q.74 $\log_2(3) \cdot \log_8(27) + 4 \cdot \log_2(2) =$ _____.

- A.** 9 **B.** 10
C. 11 **D.** 12

STATISTICS — Q.75

Q.75 If the median of $19, 15, 12, k, 8, 3$ (arranged in order) is 11 , then $k =$ _____.

- A.** 8 **B.** 9
C. 10 **D.** 11

COMPREHENSION OF UNSEEN PASSAGE — Q.76 TO Q.80

Read the following passage carefully and answer Q.76 to Q.80: The Internet is a global system of interconnected computer networks that uses the standard Internet Protocol suite (TCP/IP) to link billions of devices worldwide. It is often described as a 'network of networks' because it consists of millions of private, public, academic, business, and government networks. The World Wide Web (WWW) — a system of interlinked hypertext documents — is one of its most famous services. The web was invented by British scientist Tim Berners-Lee in 1989 while working at CERN in Geneva. Another major service is email, which allows individuals to exchange messages electronically. Today, the Internet serves billions of users and has fundamentally transformed commerce, communication, education, and entertainment across the globe.

Q.76 Who invented the World Wide Web?

- A.** Bill Gates **B.** Tim Berners-Lee
C. Steve Jobs **D.** Mark Zuckerberg

Q.77 The Internet uses which standard protocol suite?

- A.** HTML **B.** HTTP
C. TCP/IP **D.** FTP

Q.78 Where was Tim Berners-Lee working when he invented the World Wide Web?

- A.** MIT, USA
- B.** CERN, Geneva
- C.** Oxford University, UK
- D.** NASA, USA

Q.79 The Internet is best described as a _____.

- A.** Single large computer
- B.** Network of networks
- C.** Private government network
- D.** Only an academic network

Q.80 According to the passage, which of the following fields has NOT been fundamentally transformed by the Internet?

- A.** Commerce
- B.** Communication
- C.** Agriculture
- D.** Entertainment

THEORY OF COMMUNICATION — Q.81 TO Q.85

Q.81 Verbal communication can be in _____ form.

- A.** Written only
- B.** Oral and written
- C.** Oral only
- D.** None of these

Q.82 Tone of voice, voice quality, stress, and intonation are part of _____.

- A.** Body language
- B.** Sign language
- C.** Paralanguage
- D.** Code language

Q.83 A hindrance at any stage of communication that disturbs the process is known as _____.

- A.** Barrier
- B.** Sender
- C.** Receiver
- D.** Channel

Q.84 Our physical appearance is an example of _____ communication.

- A.** Oral
- B.** Written
- C.** Verbal
- D.** Non-verbal

Q.85 The process of communication is a _____ process.

- A.** One-way
- B.** Two-way
- C.** Three-way
- D.** Single-way

TECHNIQUES OF WRITING — Q.86 TO Q.90

Q.86 What is written at the top-left of a business letter?

- A.** Sender's address
- B.** Receiver's address
- C.** Date
- D.** Salutation

Q.87 A letter is incomplete and worthless without _____.

- A.** Signature
- B.** Postscript
- C.** Enclosure
- D.** Photo

Q.88 Which of the following is NOT a part of a business letter?

- A.** Photo
- B.** Salutation
- C.** Date
- D.** Signature

Q.89 The full form of C.O.D. is _____.

- A.** Cash on Delay
- B.** Cash on Delivery
- C.** Cash on Order
- D.** Cash on Distance

Q.90 Where is the salutation placed in a business letter?

- A.** Just above the date
- B.** Just below the date and receiver's address
- C.** Along with the subject line
- D.** At the end of the letter

GRAMMAR — Q.91 TO Q.95

Q.91 "Hurrah! We won the match." Identify the interjection.

- A.** Won
- B.** Match
- C.** Hurrah!
- D.** The whole sentence

Q.92 The Sun _____ in the East. Select the appropriate option.

- A.** raise
- B.** rises
- C.** rise
- D.** raised

Q.93 Your family _____ a happy family. Select the appropriate option.

- A.** are
- B.** is
- C.** has
- D.** have

Q.94 The dog is sitting _____ the cot. Select the appropriate option.

- A.** under
- B.** into
- C.** between
- D.** up

Q.95 Do you have _____ doubt in this question? Select the appropriate option.

- A.** some
- B.** any
- C.** many
- D.** much

CORRECTION OF WORDS & SENTENCES — Q.96 TO Q.100

Q.96 Select the correct spelling.

- A.** Accomodation
- B.** Accommodation
- C.** Acomodation
- D.** Acommodation

Q.97 Select the correct spelling.

- A.** Enterpeneur
- B.** Entrepreneur
- C.** Enterpreneur
- D.** Entrepeneur

Q.98 Choose the correct grammatical sentence.

- A.** Neither of the students have completed their work.
- B.** Neither of the students has completed his work.
- C.** Neither of the students were completing their work.
- D.** Neither of the student have completed work.

Q.99 Choose the correct grammatical sentence.

- A.** The news are shocking.
- B.** The news is shocking.
- C.** The news were shocking.
- D.** The news have been shocking.

Q.100 Choose the correct grammatical sentence.

A. The ship was wrecked, and every man, woman and child have drowned.

C. The ship was wrecked, and every man, woman and child is drowned.

B. The ship was wrecked, and every man, woman and child are drowned.

D. The ship was wrecked, and every man, woman and child was drowned.